STT 861: Theory of Probability & Statistics I

Chapter 4.3–4.4: Bivariate Transformations, Hierarchical Models and Mixture Distributions

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Reference: Casella & Berger Ch. 4, Sections 4.3 and 4.4

Roadmap: building new distributions from old ones

Chapter 2 asked: given X, what is the law of $g(X)$? Today we do the same for **pairs**, and then build complicated models in **layers**.

- **§4.3** $(X, Y) \mapsto (U, V)$: sums, ratios, polar coordinates, Box–Muller.
- **§4.4** $X \mid Y$ and Y: hierarchical models, mixtures, $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X|Y)]$, and $\operatorname{Var} X = \mathbb{E}\operatorname{Var}(X|Y) + \operatorname{Var}\mathbb{E}(X|Y)$.

Roadmap of Chapter 4

- 4.1–4.2 Joint, conditional, independence
- 4.3 Bivariate transformations ← today
- 4.4 Hierarchical & mixture models ← today
- 4.5 Covariance & correlation
- 4.6–4.7 Multivariate; inequalities

Motivation: what is the law of $X + Y$?

- Total claims = claim 1 + claim 2. Portfolio return = $w_1 R_1 + w_2 R_2$. Waiting time for two events = $T_1 + T_2$.
- A ratio X/Y appears in every t and F statistic; $X/(X+Y)$ is a proportion.
- Simulation: how does a computer turn two **uniform** numbers into **normal** ones?

Think (1 min)

X, Y iid $N(0,1)$. We know $X + Y \sim N(0,2)$ from MGFs. What is the *joint* law of $(X+Y,\ X-Y)$? Are they related?

Strategy: pick a second function V so $(X, Y) \leftrightarrow (U, V)$ is invertible, find the joint law of (U, V), then **marginalize** to get U.

Discrete case: add up the preimage

Discrete transformation

If (X, Y) is discrete and $(U, V) = (g_1(X, Y), g_2(X, Y))$, then

$$f_{U,V}(u, v) = \sum_{(x,y):\, g_1=u,\, g_2=v} f_{X,Y}(x, y).$$

If the map is one-to-one this is just $f_{X,Y}(h_1(u, v), h_2(u, v))$.

Example: sum of independent Poissons

$X \sim \mathrm{Poi}(\theta)$, $Y \sim \mathrm{Poi}(\lambda)$ independent. Take $U = X + Y$, $V = Y$, so $x = u - v$, $y = v$, with $0 \le v \le u$:

$$f_{U,V}(u, v) = \frac{e^{-\theta}\theta^{u-v}}{(u-v)!}\frac{e^{-\lambda}\lambda^{v}}{v!} \;\Rightarrow\; f_U(u) = \frac{e^{-(\theta+\lambda)}}{u!}\sum_{v=0}^{u}\binom{u}{v}\theta^{u-v}\lambda^{v} = \frac{e^{-(\theta+\lambda)}(\theta+\lambda)^{u}}{u!}.$$

So $X + Y \sim \mathrm{Poi}(\theta + \lambda)$. Check: $\theta = 2, \lambda = 3$ gives $P(U = 3) = e^{-5}5^3/3! \approx 0.1404$.

Continuous case: the Jacobian

Theorem (change of variables)

Let (X, Y) have joint pdf $f_{X,Y}$ and $(U, V) = (g_1(X, Y), g_2(X, Y))$ be one-to-one from $\mathcal{A} = \{f_{X,Y} > 0\}$ onto $\mathcal{B}$, with inverse $x = h_1(u, v),\ y = h_2(u, v)$. Then

$$f_{U,V}(u, v) = f_{X,Y}\big(h_1(u, v), h_2(u, v)\big)\,|J|, \quad J = \det \begin{pmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{pmatrix},$$

for $(u, v) \in \mathcal{B}$ (and 0 otherwise). J is the Jacobian of the **inverse** map.

Four-step recipe

1. Choose V and find the **support** $\mathcal{B}$ of (U, V) (draw it!).
2. Solve for x, y in terms of u, v.
3. Compute $|J|$ and substitute: $f_{U,V} = f_{X,Y}(h_1, h_2)|J|$.
4. Integrate out v: $f_U(u) = \int f_{U,V}(u, v)\, dv$.

Example 1: $U = X + Y$, $V = X - Y$ for iid $N(0,1)$

$x = \frac{u+v}{2}$, $y = \frac{u-v}{2}$, so $J = \det \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} = -\frac{1}{2}$, $|J| = \frac{1}{2}$.

$$f_{U,V}(u,v) = \frac{1}{2\pi} \exp\left\{ -\tfrac{1}{2}\left[\left(\tfrac{u+v}{2}\right)^2 + \left(\tfrac{u-v}{2}\right)^2 \right] \right\} \cdot \frac{1}{2} = \frac{1}{4\pi}\, e^{-u^2/4}\, e^{-v^2/4}$$

$$= \underbrace{\frac{1}{\sqrt{4\pi}} e^{-u^2/4}}_{N(0,2) \text{ pdf}} \cdot \underbrace{\frac{1}{\sqrt{4\pi}} e^{-v^2/4}}_{N(0,2) \text{ pdf}}$$

Conclusion

$X + Y$ and $X - Y$ are **independent** $N(0,2)$ variables. The support is all of $\mathbb{R}^2$, so the product form really means independence.

Remark

This is special to the normal. For iid $\mathrm{Exp}(1)$, $X + Y$ and $X - Y$ are *not* independent (Activity 1).

Example 2: ratio of normals is Cauchy

X, Y iid $N(0,1)$. Let $U = X/Y$ and $V = Y$ (one-to-one on $\{y \neq 0\}$, which has probability 1).

- Inverse: $x = uv,\ y = v$, so $|J| = |v|$.
- $f_{X,Y}(uv, v) = \frac{1}{2\pi} e^{-(u^2+1)v^2/2}$, and we integrate v over all of $\mathbb{R}$:

$$f_U(u) = \int_{-\infty}^{\infty} \frac{|v|}{2\pi} e^{-(1+u^2)v^2/2}\, dv = \frac{1}{\pi} \int_0^{\infty} v\, e^{-(1+u^2)v^2/2} dv = \frac{1}{\pi(1+u^2)}.$$

Result

$X/Y \sim \mathrm{Cauchy}(0,1)$. Sanity check: $P(|X/Y| < 1) = \frac{2}{\pi} \arctan 1 = \frac{1}{2}$, matching a simulation of 10^6 pairs (0.4999).

Note: the Cauchy has no mean, so ratio statistics can have heavy tails even when numerator and denominator are nice.

Example 3: Gamma and Beta from two independent Gammas

$X \sim \text{Gamma}(a, 1)$, $Y \sim \text{Gamma}(b, 1)$ independent. Let $U = X + Y$, $V = \dfrac{X}{X+Y}$.

Inverse: $x = uv$, $y = u(1-v)$, $\mathcal{B} = \{u > 0,\ 0 < v < 1\}$, and $J = \det \begin{pmatrix} v & u \\ 1-v & -u \end{pmatrix} = -u$, so $|J| = u$.

$$
\begin{aligned}
f_{U,V}(u,v) &= \frac{(uv)^{a-1}e^{-uv}}{\Gamma(a)} \cdot \frac{(u(1-v))^{b-1}e^{-u(1-v)}}{\Gamma(b)} \cdot u \\
&= \underbrace{\frac{u^{a+b-1}e^{-u}}{\Gamma(a+b)}}_{\text{Gamma}(a+b,1)} \cdot \underbrace{\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} v^{a-1}(1-v)^{b-1}}_{\text{Beta}(a,b)}
\end{aligned}
$$

Result

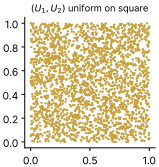
$X + Y \sim \text{Gamma}(a+b, 1)$ and $\dfrac{X}{X+Y} \sim \text{Beta}(a, b)$, and they are **independent**. (Simulation, $a = 2, b = 3$: mean of V is $0.3999 \approx 2/5$, correlation ≈ 0.)

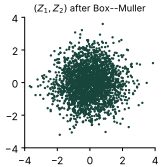
Example 4: polar coordinates and Box–Muller

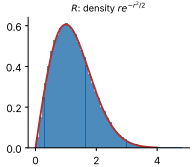
For iid $N(0,1)$ write $X = R\cos\Theta$, $Y = R\sin\Theta$; $|J| = r$. Then

$$f_{R,\Theta}(r,\theta) = \frac{1}{2\pi} e^{-r^2/2}\, r.$$

So $\Theta \sim \mathrm{Unif}(0, 2\pi)$ independent of R with density $re^{-r^2/2}$, i.e. $R^2 \sim \mathrm{Exp}(\text{mean } 2)$.







Box–Muller

Run it backwards: if U_1, U_2 iid $\mathrm{Unif}(0,1)$,

$$Z_1 = \sqrt{-2\ln U_1}\cos(2\pi U_2)$$
$$Z_2 = \sqrt{-2\ln U_1}\sin(2\pi U_2)$$

are iid $N(0,1)$.

Simulated with 10^6 draws: $\mathrm{var}(Z_1) \approx 0.996$, $\mathrm{var}(Z_2) \approx 0.999$, $\mathrm{corr}(Z_1, Z_2) \approx -0.001$.

When the map is not one-to-one

Piecewise version

Suppose $\mathcal{A}$ splits into $A_0, A_1, \ldots, A_k$ where A_0 has probability 0 and on each A_i the map is one-to-one onto $\mathcal{B}$ with inverse (h_{1i}, h_{2i}) and Jacobian J_i. Then

$$f_{U,V}(u,v) = \sum_{i=1}^{k} f_{X,Y}\big(h_{1i}(u,v), h_{2i}(u,v)\big)\,|J_i|.$$

- Typical case: $U = X^2 + Y^2$ or $U = X/Y$ with a sign ambiguity; split by sign.
- The earlier Cauchy computation is exactly this with $A_1 = \{y > 0\}$, $A_2 = \{y < 0\}$.
- Always check: does $\iint f_{U,V} = 1$?

Checklist of common results

$\mathrm{Poi}(\theta) + \mathrm{Poi}(\lambda)$	$\mathrm{Poi}(\theta + \lambda)$
$N(\mu_1, \sigma_1^2) + N(\mu_2, \sigma_2^2)$	$N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
$\mathrm{Gamma}(a, \beta) + \mathrm{Gamma}(b, \beta)$	$\mathrm{Gamma}(a + b, \beta)$
$N(0,1)/N(0,1)$	Cauchy

In-class Activity 1

Activity 1

X, Y iid $\mathrm{Exp}(1)$. Let $U = X + Y$ and $V = X - Y$.

(b) Find the inverse map and $|J|$.

(b) Find the support of (U, V) and the joint pdf $f_{U,V}$.

(c) Find the marginals of U and V. Are U, V independent?

(d) Compute $P(V > 1)$ and $P(U > 3)$.

Hint for (b): $x > 0, y > 0$ translates into a condition relating $|v|$ and u.

Activity 1: solutions

(a) $x = \frac{u+v}{2},\ y = \frac{u-v}{2}$, $|J| = \frac{1}{2}$.
(b) $x > 0, y > 0 \iff u > |v|$. So $\mathcal{B} = \{(u, v) : |v| < u\}$ and

$$f_{U,V}(u, v) = e^{-(x+y)} \cdot \tfrac{1}{2} = \tfrac{1}{2} e^{-u}, \qquad |v| < u.$$

(c) $f_U(u) = \int_{-u}^{u} \frac{1}{2} e^{-u} dv = u e^{-u}$, i.e. $U \sim \mathrm{Gamma}(2, 1)$.
$f_V(v) = \int_{|v|}^{\infty} \frac{1}{2} e^{-u} du = \frac{1}{2} e^{-|v|}$: V is double exponential (variance 2).
The support $|v| < u$ is not a rectangle, and $f_{U,V} \neq f_U f_V$, so U, V are **not independent**.
(d) $P(V > 1) = \int_1^{\infty} \frac{1}{2} e^{-v} dv = \frac{e^{-1}}{2} \approx 0.1839$. $P(U > 3) = \int_3^{\infty} u e^{-u} du = 4e^{-3} \approx 0.1991$.

Verified by simulation

10^6 draws gave $\mathrm{var}(V) = 1.994$, $P(V > 1) = 0.1842$, $P(U > 3) = 0.1992$.

Motivation: models in layers

- An insect lays Y eggs (random), and each survives independently: the number of survivors depends on Y.
- Claim counts differ across policyholders: each driver has their own rate Λ, and counts given Λ are Poisson.
- A population is a blend of two subgroups.

Hierarchy

$$Y \sim \text{(some law)}$$
$$\downarrow$$
$$X \mid Y \sim \text{(law depending on } Y\text{)}$$

Specify the **conditional** and the **marginal of the parent**, get the joint: $f(x, y) = f(x|y)f(y)$.

Goal: find the marginal law of X, and its mean and variance, without always computing the integral.

Definitions

Hierarchical model

A model specified in stages: $X \mid Y \sim f(x \mid y)$, $Y \sim f(y)$, possibly with further levels ($Y \mid \Lambda$, $\Lambda \sim \cdots$). The marginal is

$$f_X(x) = \int f(x \mid y) f(y)\, dy \quad (\text{or } \textstyle\sum_y).$$

Mixture distribution

X has a **mixture** distribution if its law arises from a hierarchy in which the parent Y is a "mixing" variable, e.g. a finite mixture $f_X = \sum_k p_k f_k$ with $p_k \geq 0$, $\sum p_k = 1$ (pick component k w.p. p_k, then draw from f_k).

- The marginal of X is often a **named** family even though $X \mid Y$ and Y are different families.
- Mixing makes X more spread out than any single component would suggest.

Example: egg-laying (Poisson–Binomial)

$Y \sim \text{Poisson}(\lambda)$ eggs; each hatches independently w.p. p: $X \mid Y \sim \text{Bin}(Y, p)$.

$$P(X = x) = \sum_{y=x}^{\infty} \binom{y}{x} p^x (1-p)^{y-x} \frac{e^{-\lambda}\lambda^y}{y!} = \frac{e^{-\lambda}(\lambda p)^x}{x!} \sum_{t=0}^{\infty} \frac{(\lambda(1-p))^t}{t!} \quad (t = y - x)$$

$$= \frac{e^{-\lambda p}(\lambda p)^x}{x!}.$$

Result ("Poisson thinning")

$X \sim \text{Poisson}(\lambda p)$. With $\lambda = 10,\ p = 0.3$: $X \sim \text{Poi}(3)$ and $P(X = 0) = e^{-3} \approx 0.0498$.

Simulation: mean 3.000, variance 2.999, $P(X = 0) = 0.0496$. **Check the intuition:** $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X|Y)] = \mathbb{E}[pY] = p\lambda$.

Law of total expectation and total variance

Theorem

$$\mathbb{E}X = \mathbb{E}\big[\mathbb{E}(X \mid Y)\big], \qquad \operatorname{Var} X = \mathbb{E}\big[\operatorname{Var}(X \mid Y)\big] + \operatorname{Var}\big(\mathbb{E}(X \mid Y)\big).$$

Proof. The first: $\mathbb{E}[\mathbb{E}(X|Y)] = \int \big(\int x f(x|y) dx \big) f(y) dy = \iint x f(x,y) dx dy = \mathbb{E}X$.

For the second, write $X - \mu = (X - \mathbb{E}(X|Y)) + (\mathbb{E}(X|Y) - \mu)$ with $\mu = \mathbb{E}X$. Square and take $\mathbb{E}$:

$$\operatorname{Var} X = \mathbb{E}[\operatorname{Var}(X|Y)] + \operatorname{Var}(\mathbb{E}(X|Y)) + 2\,\mathbb{E}\big[(X - \mathbb{E}(X|Y))(\mathbb{E}(X|Y) - \mu)\big],$$

and the cross term is 0 because conditionally on Y the second factor is a constant and $\mathbb{E}[X - \mathbb{E}(X|Y) \mid Y] = 0$. □

In words

Total variance = **average within-group variance** + **variance of group means**. Hence $\operatorname{Var} X \geq \mathbb{E} \operatorname{Var}(X|Y)$: mixing never reduces spread.

Example: a random sum (compound Poisson)

$N \sim \mathrm{Poisson}(20)$ claims; claim sizes $X_1, X_2, \ldots$ iid with mean 5, variance 4, independent of N. Total $S = \sum_{i=1}^{N} X_i$.

$$\mathbb{E}(S \mid N) = 5N, \quad \mathrm{Var}(S \mid N) = 4N.$$
$$\mathbb{E}S = \mathbb{E}[5N] = 5 \cdot 20 = 100.$$
$$\mathrm{Var}\, S = \mathbb{E}[4N] + \mathrm{Var}(5N) = 4 \cdot 20 + 25 \cdot 20 = 580.$$

General formula

$\mathbb{E}S = \mathbb{E}N \cdot \mu$, $\mathrm{Var}\, S = \mathbb{E}N \cdot \sigma^2 + \mathrm{Var}\, N \cdot \mu^2$.

- Without conditioning you would need the whole law of S; with it, two lines.
- Simulation ($X_i \sim N(5, 2^2)$): $\mathbb{E}S \approx 99.9$, $\mathrm{Var}\, S \approx 580.8$.

Example: Gamma–Poisson gives the Negative Binomial

$Y \mid \Lambda \sim \text{Poisson}(\Lambda)$, $\Lambda \sim \text{Gamma}(\alpha, \beta)$ (shape α, scale β).

$$P(Y=y) = \int_0^\infty \frac{e^{-\lambda}\lambda^y}{y!} \frac{\lambda^{\alpha-1}e^{-\lambda/\beta}}{\Gamma(\alpha)\beta^\alpha} d\lambda = \binom{y+\alpha-1}{y} \left(\frac{1}{1+\beta}\right)^\alpha \left(\frac{\beta}{1+\beta}\right)^y.$$

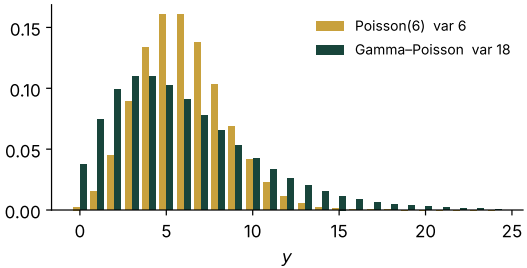
So $Y \sim \text{NegBin}\big(\alpha,\ p = \frac{1}{1+\beta}\big)$ (for integer α).

$\mathbb{E}Y = \mathbb{E}\Lambda = \alpha\beta$

$\text{Var}\, Y = \mathbb{E}\Lambda + \text{Var}\,\Lambda = \alpha\beta + \alpha\beta^2$

Overdispersion: $\text{Var}\, Y > \mathbb{E}Y$, unlike a pure Poisson.

$\alpha = 3, \beta = 2$: mean 6, variance 18,
$P(Y=0) = (1/3)^3 \approx 0.037$ (a Poisson(6) has $P(0) = 0.0025$).



Real count data (accidents, reads, claims) is almost always overdispersed, hence the Negative Binomial.

Example: Beta–Binomial

$X \mid P \sim \mathrm{Bin}(n, P)$ with $P \sim \mathrm{Beta}(\alpha, \beta)$ (success probability varies across units).

$$\mathbb{E}X = \mathbb{E}[nP] = \frac{n\alpha}{\alpha+\beta}, \qquad \mathrm{Var}\, X = \mathbb{E}[nP(1-P)] + \mathrm{Var}(nP) = \frac{n\alpha\beta(\alpha+\beta+n)}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Numbers: $n = 10$, $\alpha = 2$, $\beta = 3$

$\mathbb{E}X = 4$, $\mathrm{Var}\, X = \dfrac{10 \cdot 6 \cdot 15}{25 \cdot 6} = 6$, versus $\mathrm{Var} = 10(0.4)(0.6) = 2.4$ for a plain $\mathrm{Bin}(10, 0.4)$.
Simulation: mean 3.9996, variance 5.997.

- Same overdispersion story: heterogeneity in p inflates variance by the factor $\frac{\alpha+\beta+n}{\alpha+\beta+1} = 2.5$ here.
- The marginal is not Binomial, even though every conditional is.

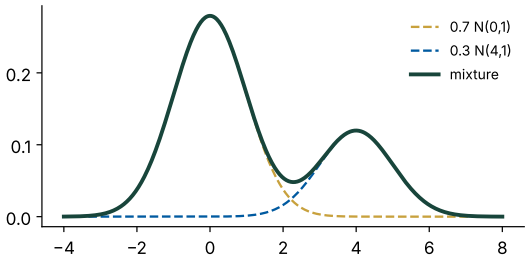
Example: finite normal mixture

A population is 70% group A, $N(0,1)$, and 30% group B, $N(4,1)$. Let $Z \in \{A, B\}$ be the label.

$$f_X(x) = 0.7\,\phi(x) + 0.3\,\phi(x-4).$$

$$\mathbb{E}X = 0.7(0) + 0.3(4) = 1.2$$

$$\operatorname{Var} X = \underbrace{1}_{\mathbb{E}\operatorname{Var}(X|Z)} + \underbrace{16(0.3)(0.7)}_{\operatorname{Var}\mathbb{E}(X|Z)} = 4.36$$



- The mixture is **not** normal: it is bimodal and skewed. Its mean 1.2 lies in a region of low density.
- Simulation: mean 1.2005, variance 4.364.

Example: a normal hierarchy stays normal

$X \mid Y \sim N(Y, 1)$, $Y \sim N(0, 4)$.

- $\mathbb{E}X = \mathbb{E}Y = 0$, $\operatorname{Var} X = \mathbb{E}[1] + \operatorname{Var}(Y) = 1 + 4 = 5$.
- In fact $X = Y + \varepsilon$ with independent $\varepsilon \sim N(0, 1)$, so $X \sim N(0, 5)$ exactly.
- Here the mixing distribution and the conditional are both normal, and the marginal is again normal. This is the building block of random-effects models.

Contrast

Normal mixture over a **discrete** label (previous slide): not normal.
Normal over a **normal** location: normal.

Simulation: mean 0.001, variance 4.996.

Summary of hierarchical models

$X \mid Y$	Y	Marginal of X	Var X vs. $\mathbb{E}X$
$\mathrm{Bin}(Y, p)$	$\mathrm{Poi}(\lambda)$	$\mathrm{Poi}(\lambda p)$	equal
$\mathrm{Poi}(\Lambda)$	$\Lambda \sim \mathrm{Gamma}(\alpha, \beta)$	NegBin	over-dispersed
$\mathrm{Bin}(n, P)$	$P \sim \mathrm{Beta}(\alpha, \beta)$	Beta–Binomial	over-dispersed
$N(Y, 1)$	$N(0, 4)$	$N(0, 5)$	—
$N(\mu_Z, 1)$	discrete Z	finite mixture	bimodal

Take-aways

1. Transformations: find (U, V) jointly, then marginalize; the Jacobian is that of the inverse map.
2. Hierarchies: $f(x, y) = f(x|y)f(y)$; use $\mathbb{E}X = \mathbb{E}\mathbb{E}(X|Y)$ and $\mathrm{Var}\, X = \mathbb{E}\,\mathrm{Var}(X|Y) + \mathrm{Var}\,\mathbb{E}(X|Y)$.
3. Mixing produces extra variance (overdispersion).

In-class Activity 2

Activity 2

(a) $N \sim \text{Poisson}(3)$ customers, each buys independently w.p. 0.5. Let X be the number of buyers. Find the law of X and $P(X = 0)$.

(b) $Y \mid \Lambda \sim \text{Poisson}(\Lambda)$ and $\Lambda \sim \text{Gamma}(\alpha = 2, \beta = 3)$. Find $\mathbb{E}Y$, Var Y, and $P(Y = 0)$.

(c) $X \mid Y \sim N(Y, 1)$ and $Y \sim N(0, 4)$. Find the marginal law of X.

Activity 2: solutions

(a) By Poisson thinning, $X \sim \mathrm{Poi}(3 \cdot 0.5) = \mathrm{Poi}(1.5)$, $P(X = 0) = e^{-1.5} \approx 0.2231$.

(b) $\mathbb{E}Y = \mathbb{E}\Lambda = \alpha\beta = 6$.
$\mathrm{Var}\, Y = \mathbb{E}\Lambda + \mathrm{Var}\, \Lambda = 6 + \alpha\beta^2 = 6 + 18 = 24$.
$P(Y = 0) = \mathbb{E}e^{-\Lambda} = (1 + \beta)^{-\alpha} = 4^{-2} = \frac{1}{16} = 0.0625$.

(c) $X = Y + \varepsilon$ with $\varepsilon \sim N(0, 1)$ independent of Y, so $X \sim N(0, 5)$.

Simulation check

(a) mean 1.500, $P(0) = 0.2236$. (b) mean 6.004, variance 24.00, $P(0) = 0.0623$. (c) variance 4.996.

Wrap-up and next time

Remember

- $f_{U,V} = f_{X,Y}(h_1, h_2)\,|J|$, with J from the inverse.
- Sums/ratios: Gamma–Beta pair, Cauchy, Box–Muller.
- Law of total variance.
- Hierarchies give overdispersed families.

Next: §4.5

Covariance and correlation: when two variables are not independent, how strongly are they linearly related?